

**ICPRN 2026**



International Conference on  
Pattern Recognition **LYON**

# Keynote Speakers



Henrik I. Christensen

Building a perception stack for  
autonomous driving



Alicia Fornés

Challenges in Handwriting Recognition: The case of  
Historical Texts, Music Scores and Ciphers

*"As we reach 99.9999%, we have solved the dataset, but we have not solved the problem". [Alicia Fornés](#) ICPR26.*

*"There is no single model in the current SOTA that can generalize across all benchmarks. There is no method that we can call 'The ...'."*

— [Alicia Fornés](#) [#ICPR2026](#)

# Dual the Reasoning, Double the Insight with Tambl: A Self-Supervised Framework for Skeleton Action Representation

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<sup>1</sup>*Inria*

<sup>2</sup>*Woven by Toyota*

<sup>3</sup>*Toyota Motor Europe*



**Mahmoud ALI**



**Snehashis Majhi**



**Di Yang\***



**Quan Kong**



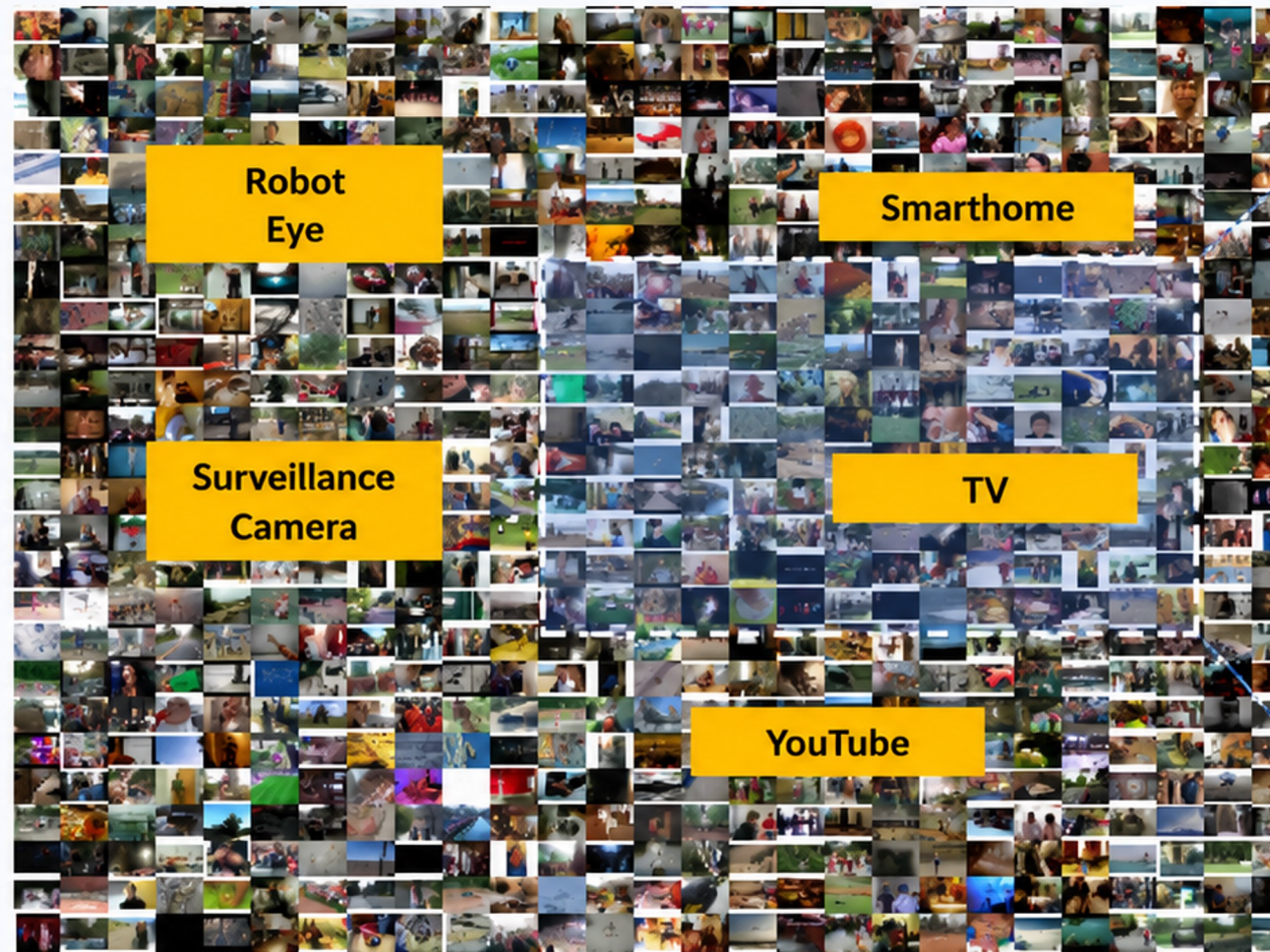
**Gianpiero Francesca**



**Francois Bremond**



## General Video (All Categories)



## Zoom to Subset (**Human Videos**)





# Human Action recognition tasks

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- Action Classification: **what?**



- Temporal Action Detection/Localization: **when?**

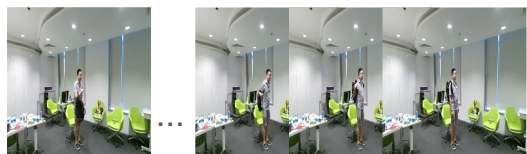


# Motivation

*To Improving skeleton motion representation that can generalize well for 3D Lab-setting dataset (e.g., NTU-RGB-D) and 2D real-world dataset (e.g., SmartHome) in various scenarios and downstream tasks.*

## Controlled Lab Setting dataset

### Lab Setting Recording

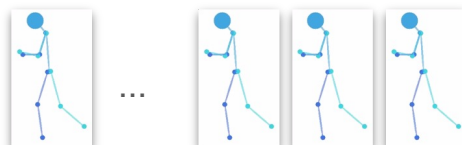


Take off bag

NTU-RGB+D 60

## Real World datasets

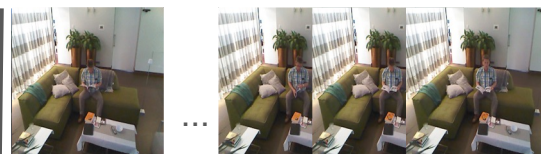
### Web Videos



Pushing car

Posetics

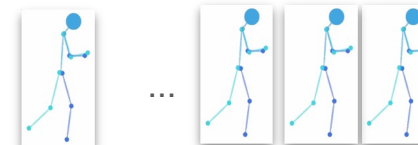
### Activity of Daily Living



Read book

SmartHome

### Sports Environment



Baseball pitch

PennAction





# Major Challenges in real-world videos

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## Spatial

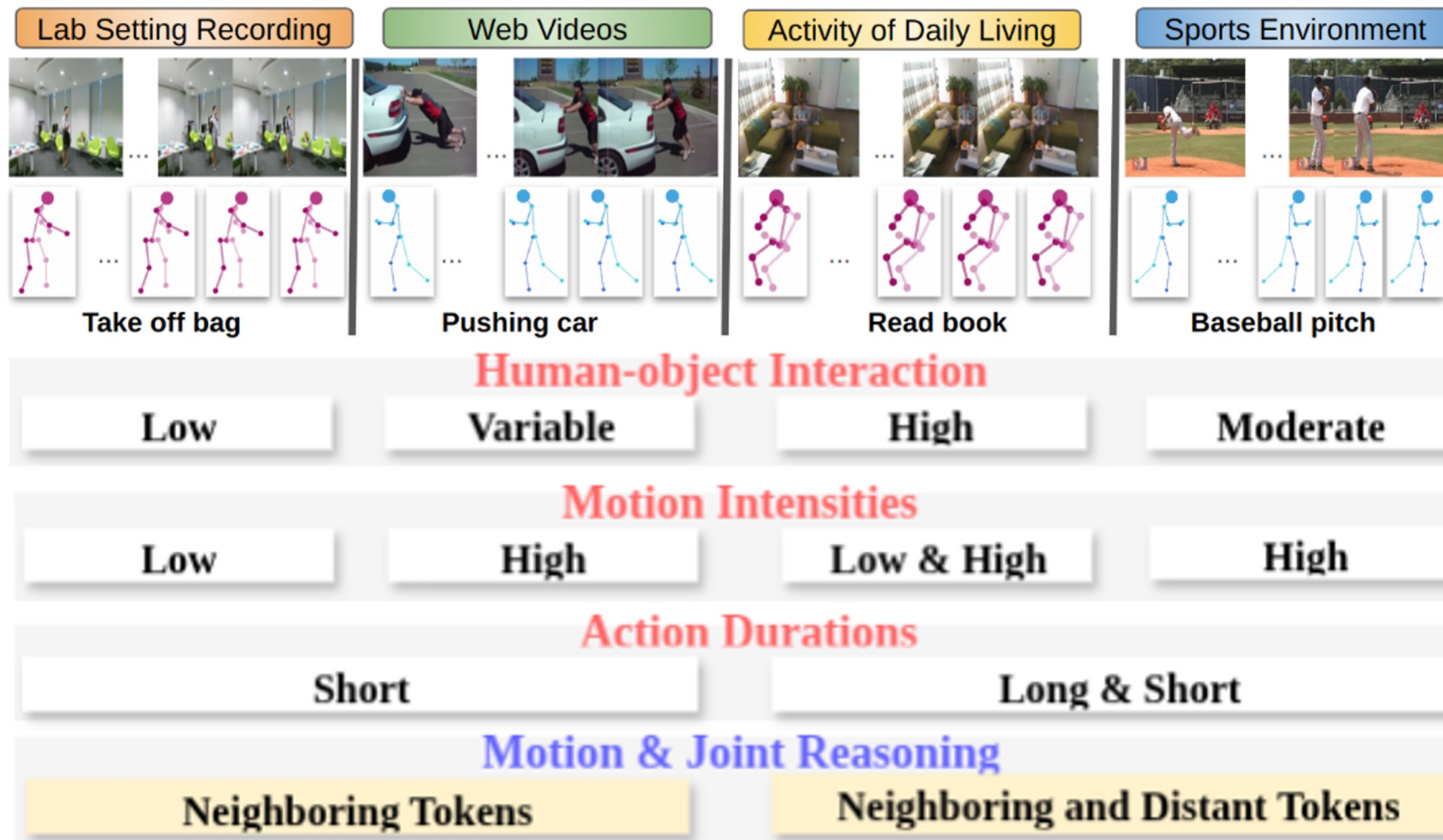
- View variance
- Occlusion
- Interaction with Objects

## Temporal

- Long-term video
- Composite activities
- Co-occurring actions



# Variability Across Multiple Datasets and Scenarios



# Background



## VLMs (Video-Language Models)

Large-scale video-language models capture global semantics, but often under-model human body motion.

### Model Characteristics

|  |                    |                                                                                  |   |
|--|--------------------|----------------------------------------------------------------------------------|---|
|  | Model Size         | Very heavy<br>(Billions of parameters)                                           | ✓ |
|  | Compute            | High<br>(Needs large GPU/TPU)                                                    | ✓ |
|  | Data               | Trained on massive video-text data<br>(web-scale)                                | ✓ |
|  | Generalization     | Good at zero-shot understanding<br>of new tasks                                  | ✓ |
|  | Human Motion       | Under-models fine-grained<br>body motion                                         | ✗ |
|  | Object Information | Rich object & scene information<br>(recognizes objects, scenes,<br>interactions) | ✓ |

VS.



## Skeleton Models

Skeleton encoders capture motion, but are usually trained in closed-set settings and transfer poorly, as well as missing object information.

### Model Characteristics

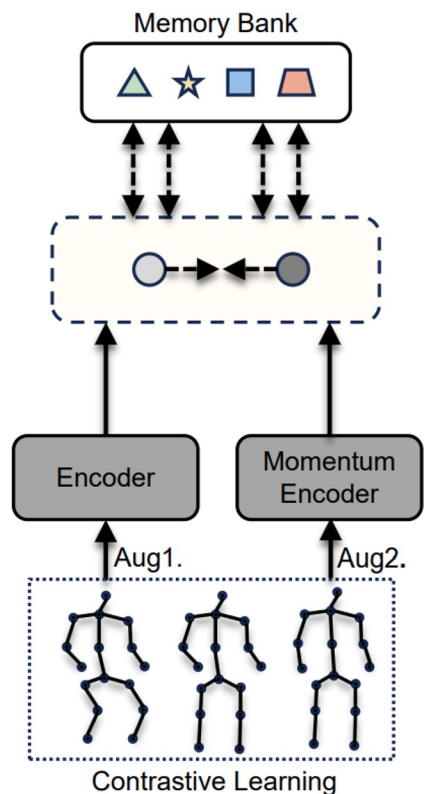
|  |                    |                                                                        |   |
|--|--------------------|------------------------------------------------------------------------|---|
|  | Model Size         | Lightweight<br>(Millions of parameters)                                | ✓ |
|  | Compute            | Low<br>(Runs on CPU / small GPU)                                       | ✓ |
|  | Data               | Trained on closed-set datasets<br>(e.g., NTU RGB+D, Kinetics-Skeleton) | ✗ |
|  | Generalization     | Poor transfer to new<br>domains / activities                           | ✗ |
|  | Human Motion       | Strong at fine-grained<br>body motion                                  | ✓ |
|  | Object Information | Missing object & scene<br>information                                  | ✗ |





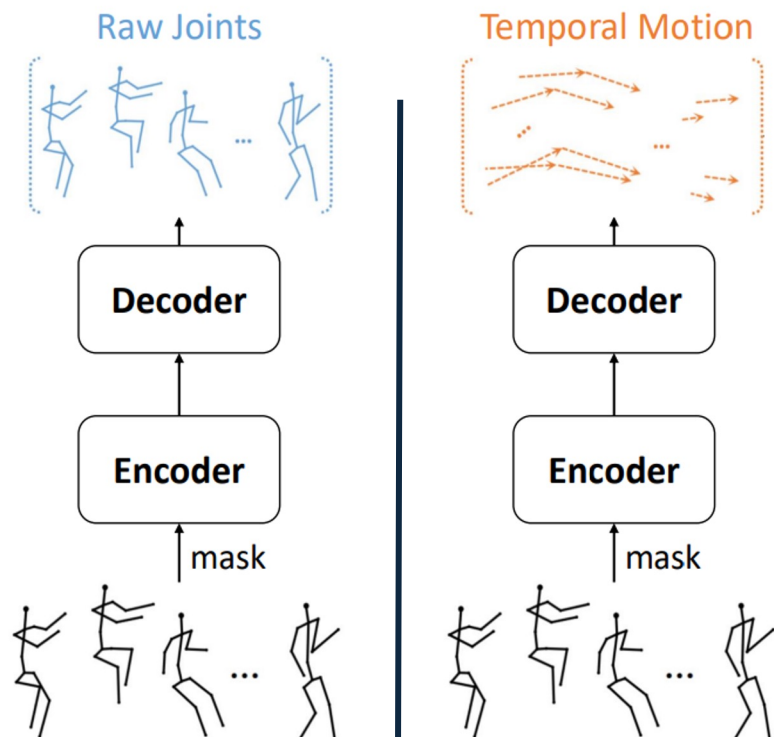
# Overview: SSL Skeleton Existing Methods

## Contrastive



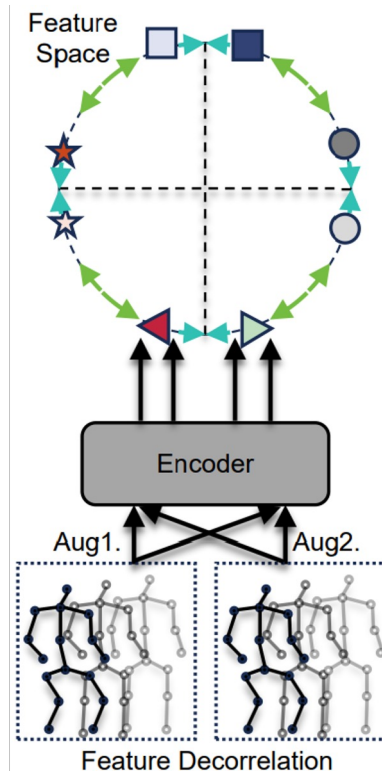
- SCDNet (AAAI 24)
- HiCo (AAAI 23)

## Generative



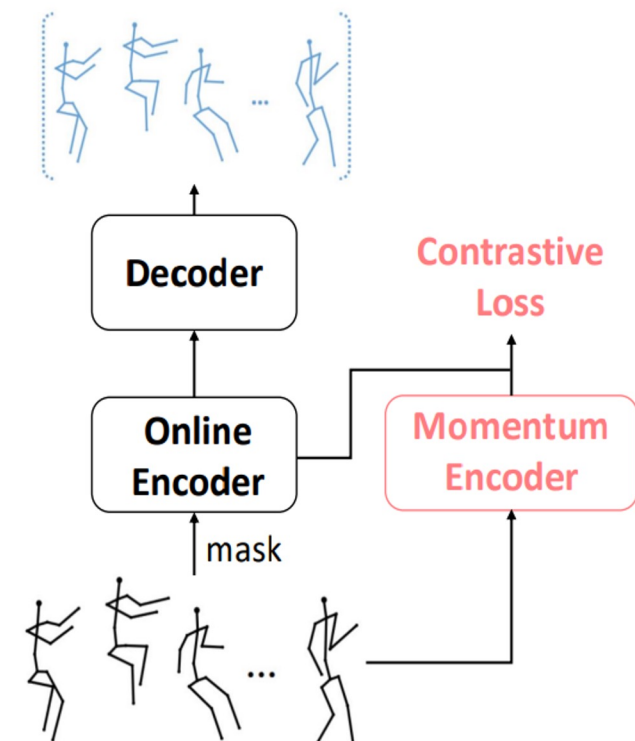
- SkeletonMAE (ICME-W 23)
- MAMP (ICCV 23)

## Decorrelation



- USDRL (AAAI 25)

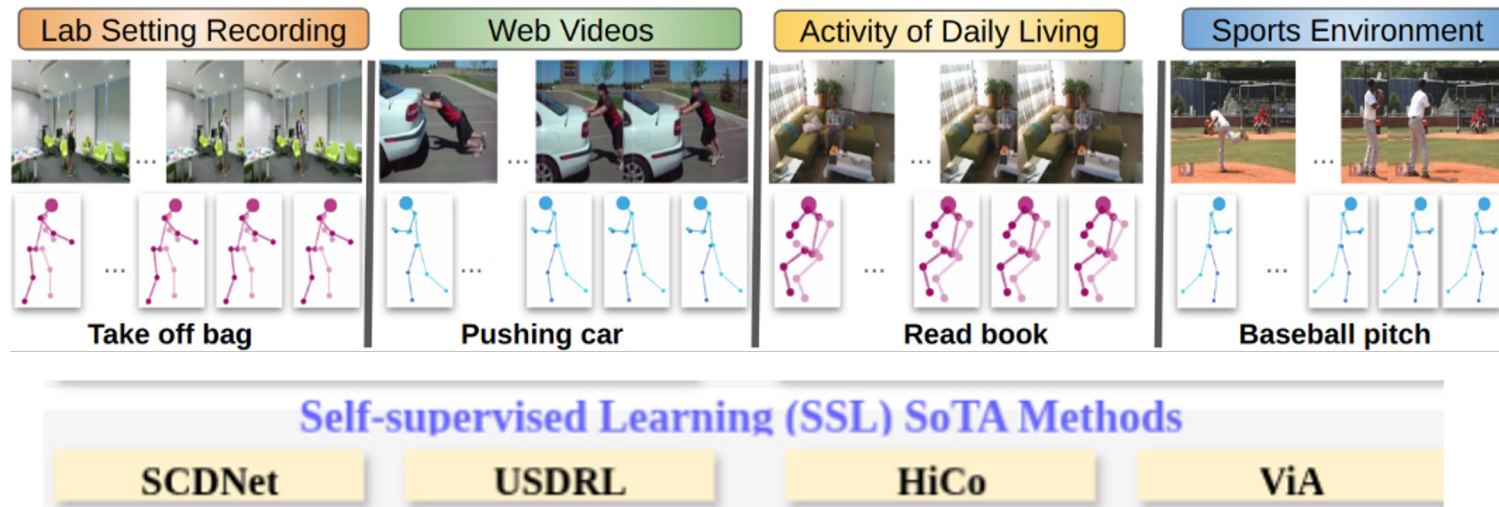
## Hybrid



- NAT (AAAI 25)
- ViA (IJCV 24)

# Limitation

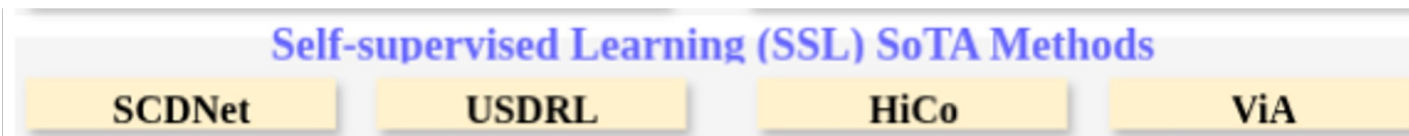
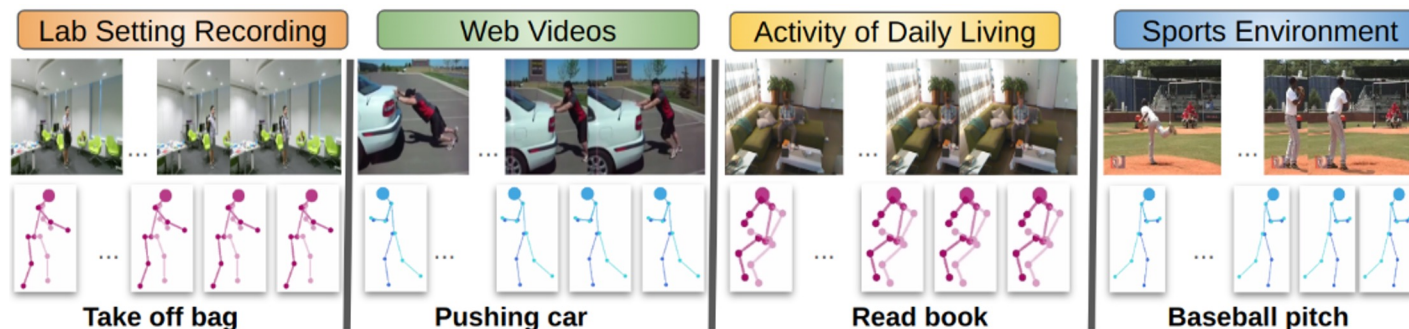
Current skeleton-based models mainly focus on controlled lab-setting datasets and struggle to generalize to real-world 2D scenarios, which involve greater diversity in motion intensities, action durations, and human object interactions.



**The Problem: No Single Model Generalizes Everywhere**



# Self-Supervised Learning SOTA methods



Can a unified SSL method out-perform individual scenario SoTA methods?

**YES, Go for our Tambl**

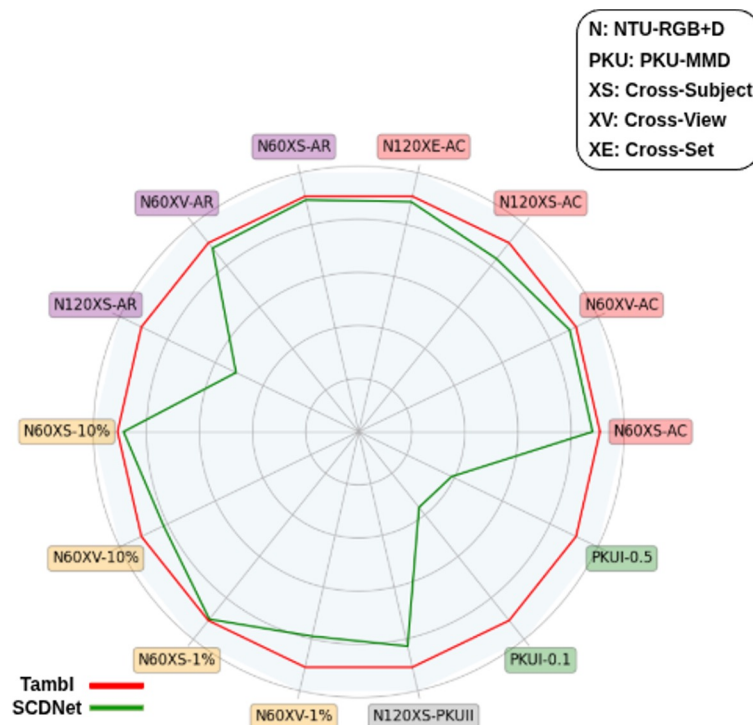




# Tambl Vs. SOTA Comparison

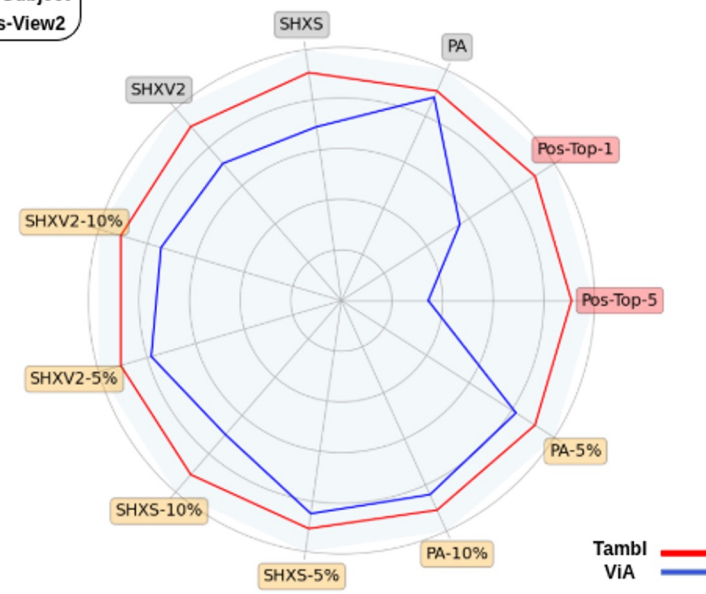
Can a unified SSL method out-perform individual scenario SoTA methods?  
**YES, GO for our Tambl**

■ Action Recognition ■ Action Retrieval ■ Transfer Learning ■ Semi-supervised ■ Action Detection



Lab Setting Datasets  
NTU-60 | NTU-120 | PKU-MMD

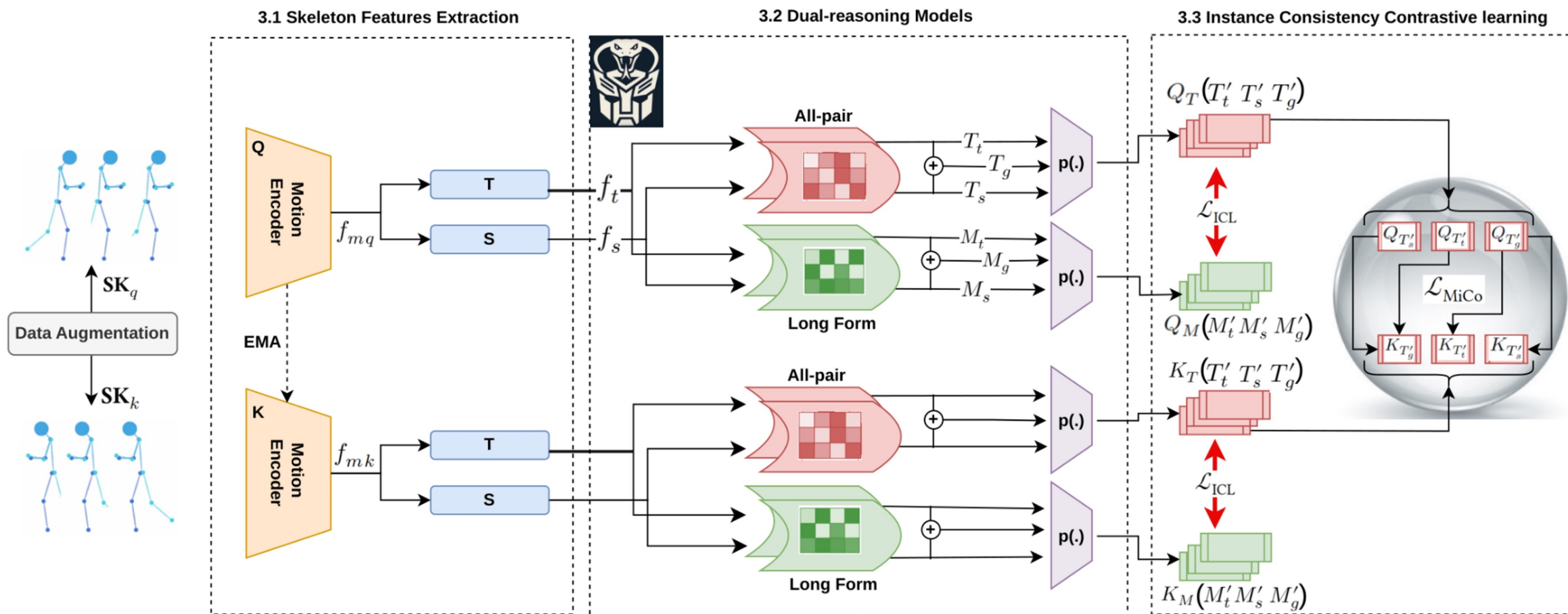
Pos: Posetics  
PA: Penn Action  
SH: Smart Home  
XS: Cross-Subject  
XV2: Cross-View2



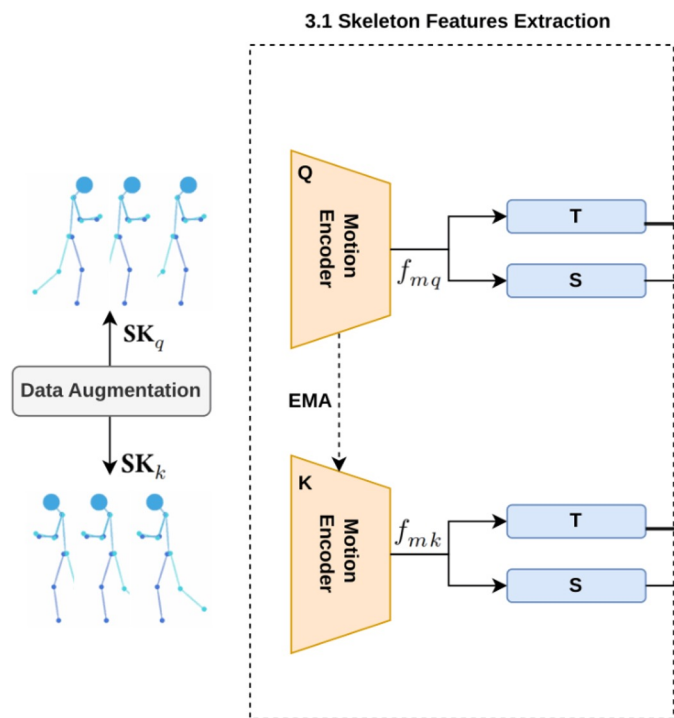
Real world Datasets  
Posetics | SmartHome | Penn Action



# Framework Overview [TambI]



# [TambI] 1. Skeleton Features Extraction



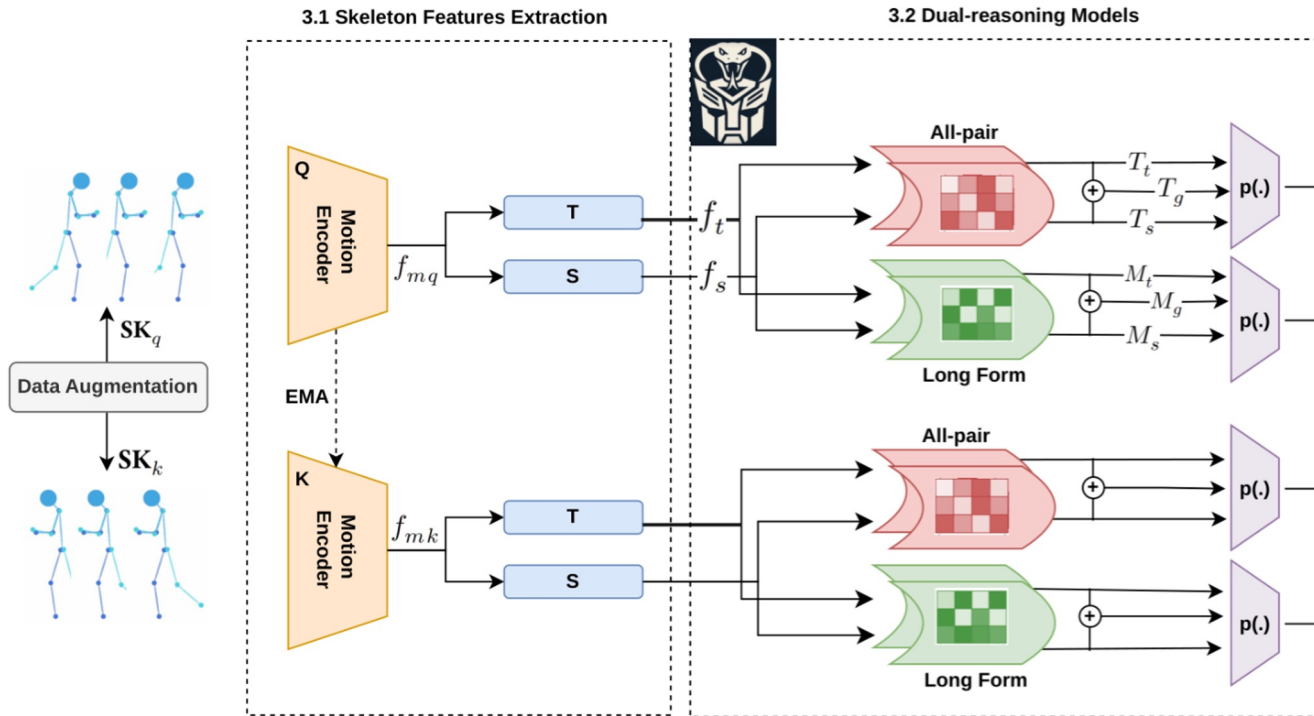
## 1. Feature Extraction

Two augmented views ( $SK_q$ ,  $SK_k$ ) pass through a CTR-GCN motion encoder (query: backprop, key: EMA momentum), then decouple into temporal (ft) and spatial (fs) features.





# [TambI] 2. Dual reasoning Models



## 1. Feature Extraction

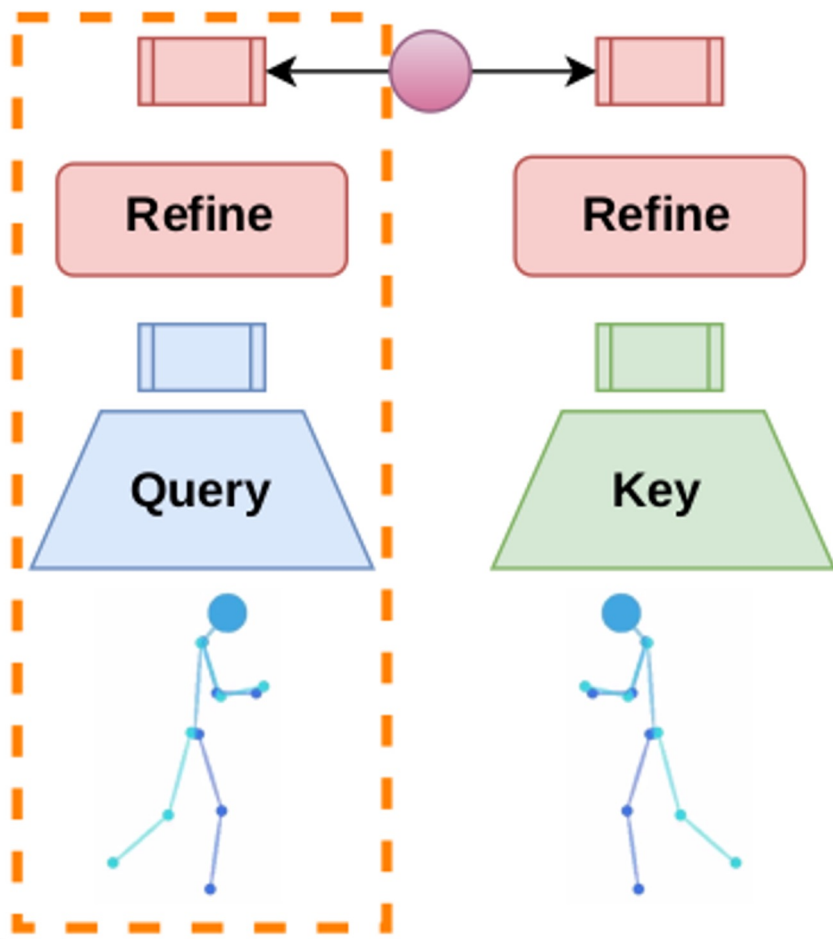
Two augmented views ( $SK_q, SK_k$ ) pass through a CTR-GCN motion encoder (query: backprop, key: EMA momentum), then decouple into temporal ( $f_t$ ) and spatial ( $f_s$ ) features.

## 2. Dual-Reasoning

Transformer (all-pair attention) and Mamba (selective state-space) each refine  $f_t$  &  $f_s$ , fused into global  $T_g$  and  $M_g$  representations.



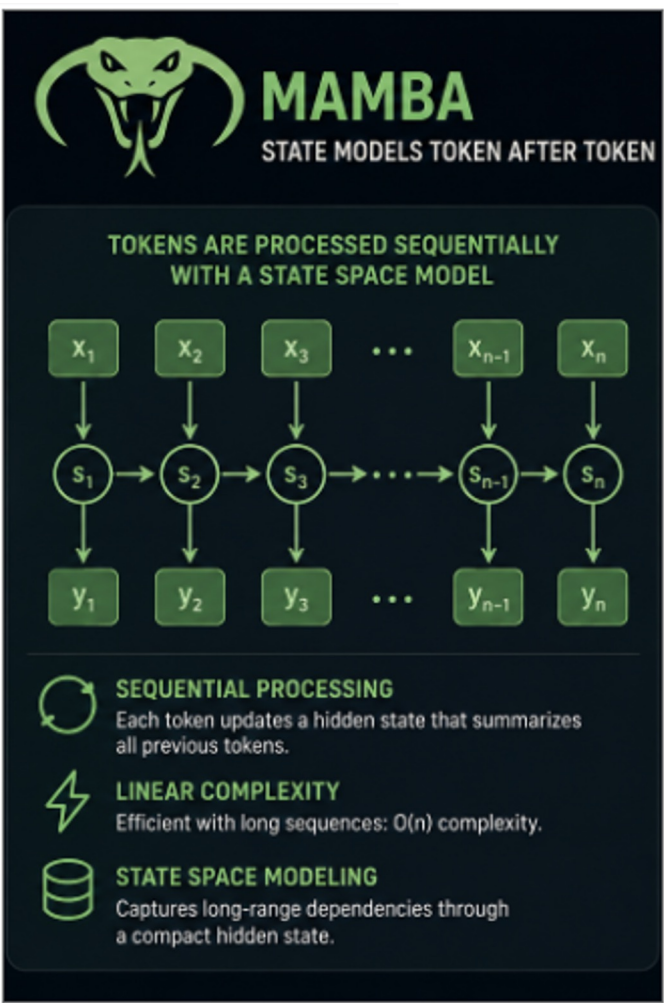
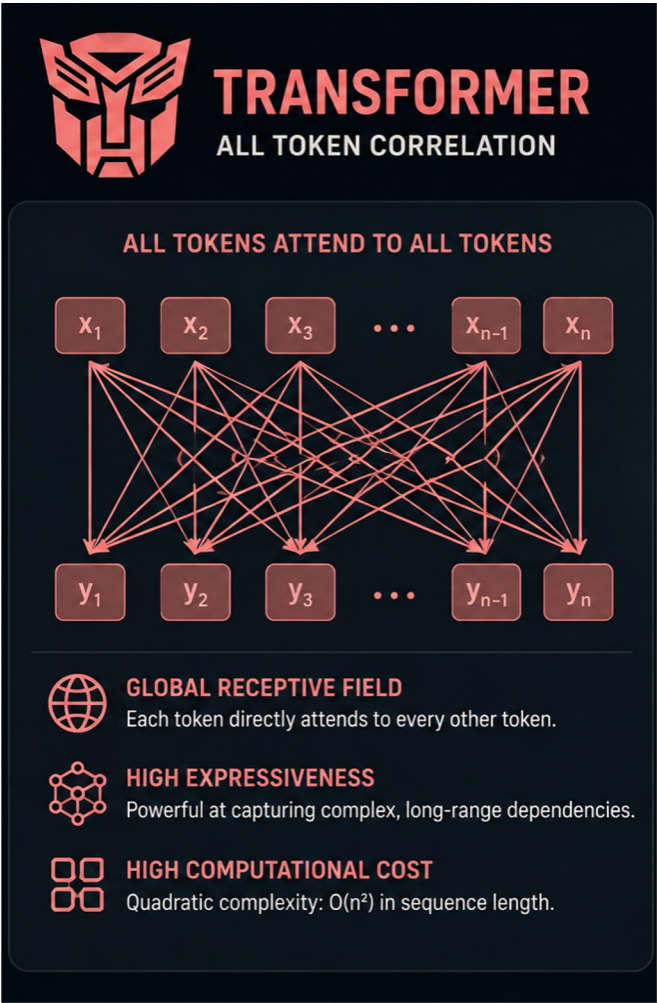
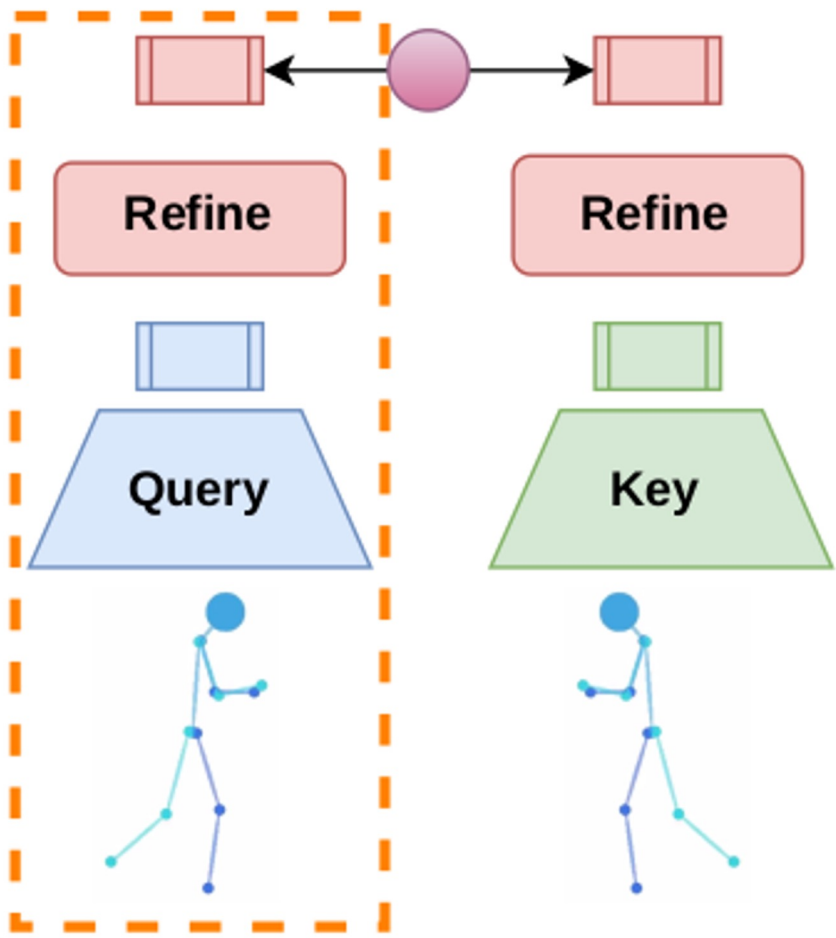
# Recall: Single reasoning SOTA



Wu, C., et al.: Scd-net: spatiotemporal clues disentanglement network for selfsupervised skeleton-based action recognition. In: AAAI (2024)



# Recall: Single reasoning SOTA



Wu, C., et al.: Scd-net: spatiotemporal clues disentanglement network for selfsupervised skeleton-based action recognition. In: AAAI (2024)





# Tambl: Dual Reasoning Models



# Tambl: Dual Reasoning Models

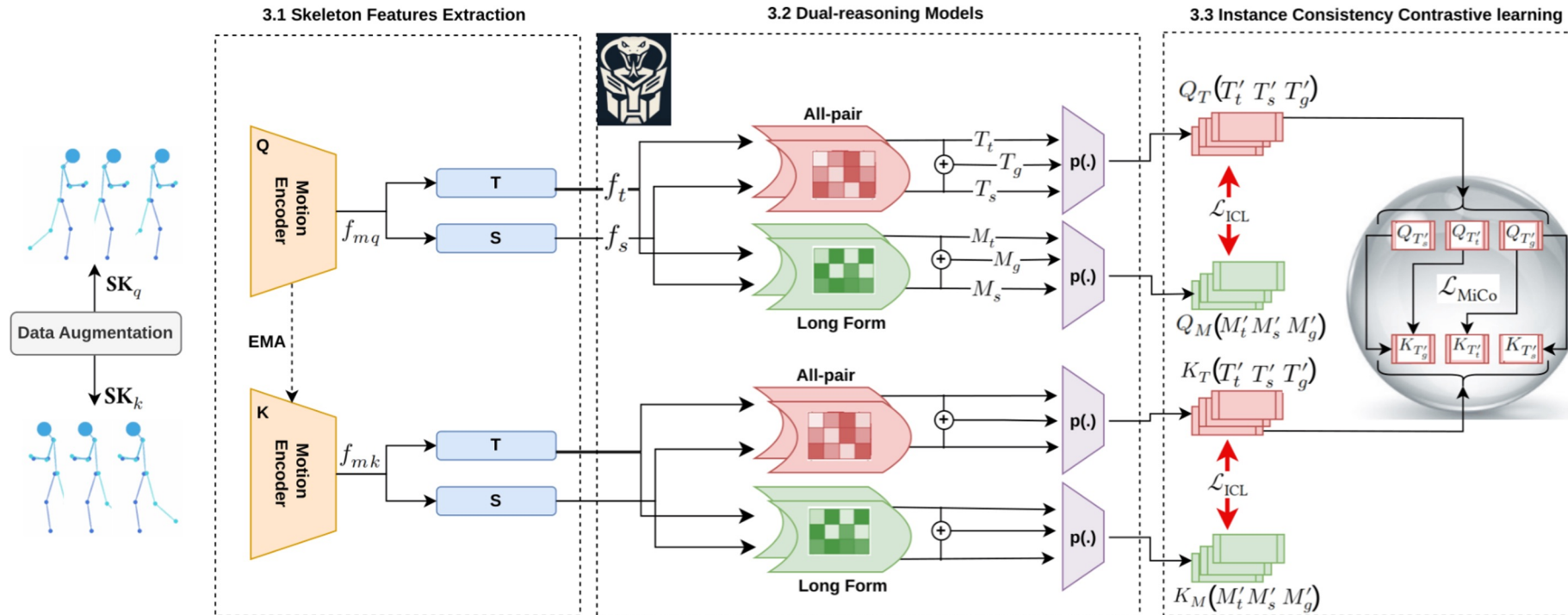


- **A Transformer** branch captures fine-grained, short-term local motion.
- **A Mamba** (state-space) branch models long-range temporal dependencies.
- **Fusing both** spans a richer hypothesis space than either alone.





# [TambI] 3. Instance Consistency Contrastive learning



## 1. Feature Extraction

Two augmented views ( $SK_q$ ,  $SK_k$ ) pass through a CTR-GCN motion encoder (query: backprop, key: EMA momentum), then decouple into temporal (ft) and spatial (fs) features.

## 2. Dual-Reasoning

Transformer (all-pair attention) and Mamba (selective state-space) each refine ft & fs, fused into global Tg and Mg representations.

## 3. Instance-Consistency Contrastive Learning

ICL aligns Transformer/Mamba embeddings; MiCo mixes spatial/ temporal/ global views for InfoNCE-style contrastive discrimination. Mamba branch is dropped at inference.





# Training objectives

## Two complementary pre-training objectives

### Objective 1: Instance Consistency Loss (ICL)

- Minimizes redundancy and noise by enforcing alignment between Transformer and Mamba features within each representation branch.
- This promotes consistent and complementary features across encoders and strengthens subsequent contrastive learning.

$$\mathcal{L}_{\text{ICL}} = \|Q_T - Q_M\|^2.$$

### Objective 2: Mixing Contrastive Loss (MiCo)

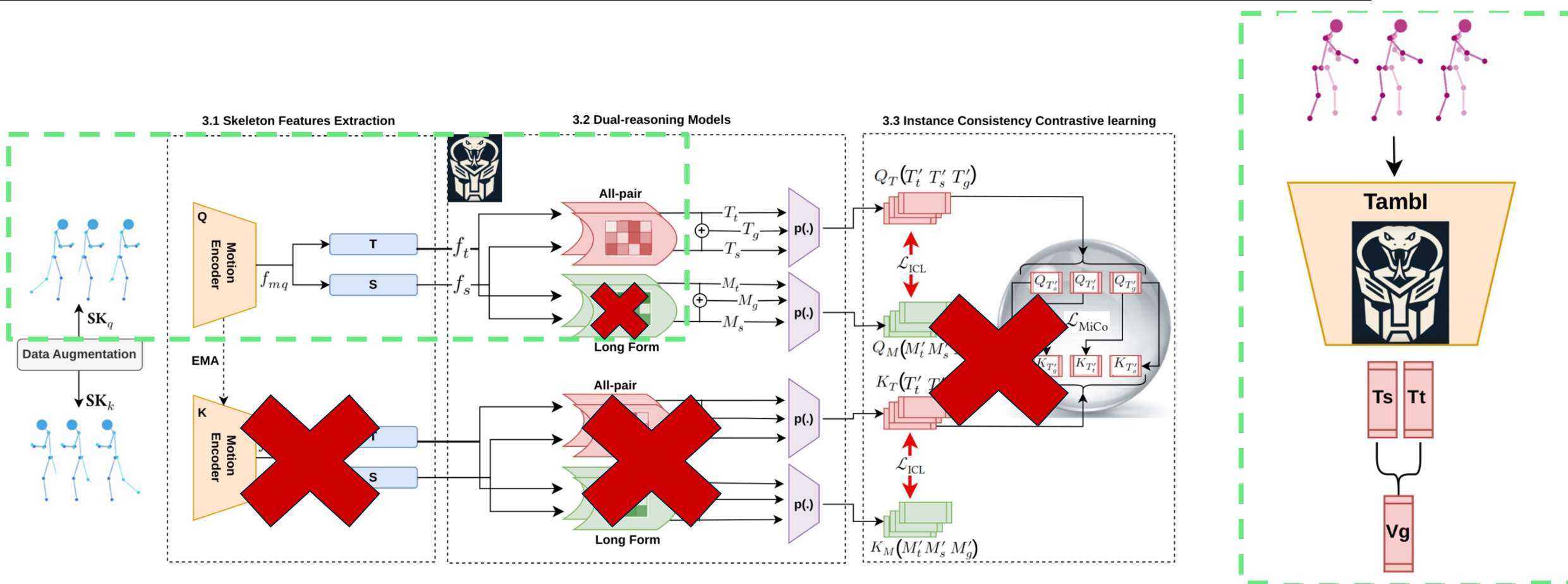
- Enhances discrimination in the embedding space by mixing spatial, temporal, and global views from different branches.
- Built on a mix InfoNCE with a memory bank of negatives (MoCo-v2 style).

$$\mathcal{L}_{\text{MiCo}} = \mathcal{L}_{\text{info}}(Q_s, K_g) + \mathcal{L}_{\text{info}}(Q_t, K_g) + \mathcal{L}_{\text{info}}(Q_g, K_s) + \mathcal{L}_{\text{info}}(Q_g, K_t)$$

$$\mathcal{L}_{\text{ICCo}} = \mathcal{L}_{\text{ICL}} + \mathcal{L}_{\text{MiCo}}$$



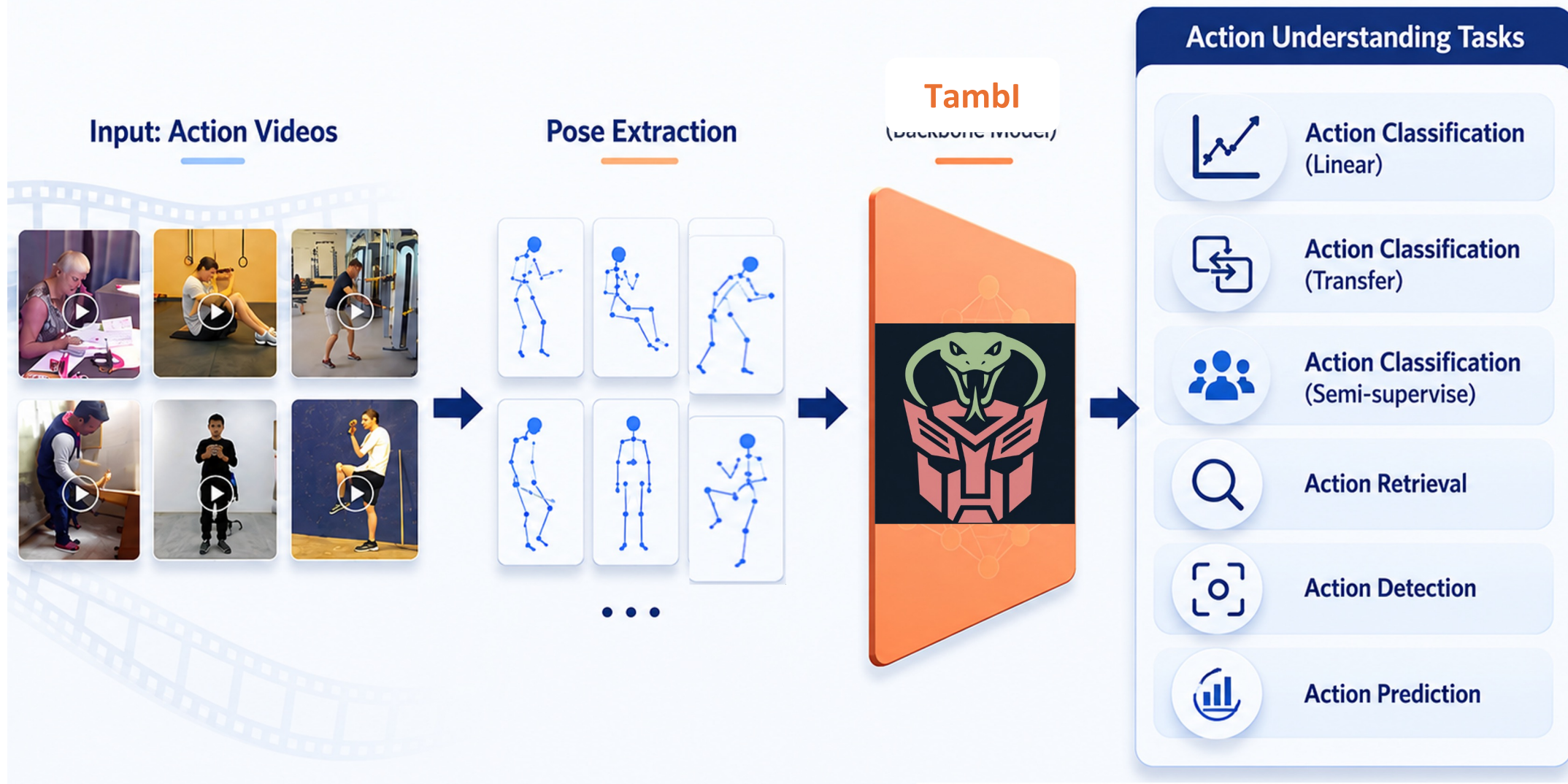
# Training Vs. Inference



One pretrained backbone (Mamba dropped at inference) achieves SoTA on 6 benchmarks spanning lab (NTU-60/120, PKU-MMD) and real-world (SmartHome, Penn Action, Posetics) settings.

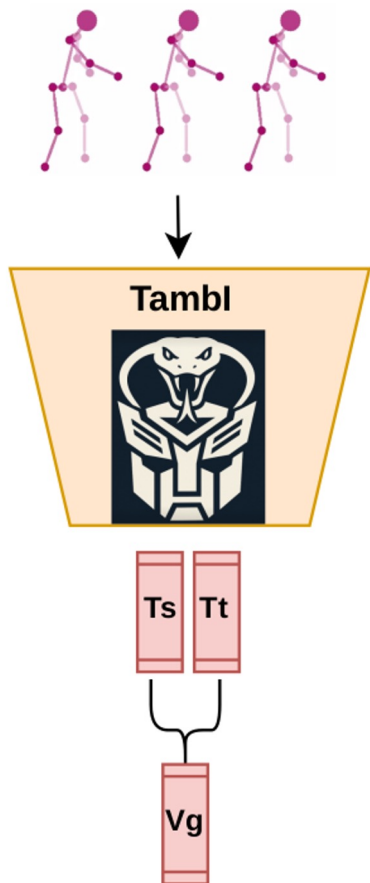


# Evaluation





# Evaluation setting



## Action Classification

### Datasets:

- Smarthome,
- Posetics,
- Penn Action.
- NTU-RGB+D 60
- NTU-RGB+D 120
- PKU-MMD II

### Evaluate:

linear probing and transfer learning.

Top-1

## Action Detection

### Datasets:

- PKU-MMD I

### Evaluate:

frame-wise multi-label action detection.

mAP

All downstream evaluations use skeleton inputs and the transferred skeleton motion encoder.



# Result [Action recognition]: 3D Lab setting datasets

| Method                            | NTU-RGB+D 60 CS(%) |             | NTU-RGB+D 60 CV(%) |             | NTU-RGB+D 120 CS(%) |             | NTU-RGB+D 120 CSet(%) |             |
|-----------------------------------|--------------------|-------------|--------------------|-------------|---------------------|-------------|-----------------------|-------------|
|                                   | J                  | J+B+M       | J                  | J+B+M       | J                   | J+B+M       | J                     | J+B+M       |
| MS <sup>2</sup> L [13] (ACMMM 20) | 52.6               | -           | -                  | -           | -                   | -           | -                     | -           |
| ISC [22] (ACMMM 21)               | 76.3               | -           | 85.2               | -           | 67.1                | -           | 67.9                  | -           |
| CrosSCLR [12] (CVPR 21)           | 72.9               | 77.8        | 79.9               | 83.4        | -                   | 67.9        | -                     | 66.7        |
| CPM [33] (ECCV 22)                | 78.7               | 83.2        | 84.9               | 87.0        | 68.7                | 73.0        | 69.6                  | 74.0        |
| CMD [18] (ECCV 22)                | 79.8               | 84.1        | 86.9               | 90.9        | 70.3                | 74.7        | 71.5                  | 76.1        |
| ActCLR [14] (CVPR 23)             | 80.9               | 84.3        | 86.7               | 88.8        | 69.0                | 74.3        | 70.5                  | 75.7        |
| AimCLR [23] (AAAI 22)             | 74.3               | 78.9        | 79.7               | 83.8        | 63.4                | 68.2        | 63.4                  | 68.8        |
| RVTCLR+ [37] (ICCV 23)            | 74.7               | 79.7        | 79.1               | 84.6        | -                   | 60.0        | -                     | 68.9        |
| HiCLR [1] (AAAI 23)               | 80.4               | -           | 85.5               | -           | 70.0                | -           | 70.4                  | -           |
| UmURL [21] (ACMMM 23)             | 82.3               | 84.2        | 89.8               | 90.0        | 73.5                | 75.2        | 74.3                  | 76.3        |
| PCM <sup>3</sup> [35] (ACMMM 23)  | 83.9               | 87.4        | 90.4               | 93.1        | 76.5                | 80.0        | 77.5                  | 81.2        |
| MAMP [17] (ICCV 23)               | 84.9               | -           | 89.1               | -           | 78.6                | -           | 79.1                  | -           |
| HiCo [5] (AAAI 23)                | 81.1               | 83.8        | 88.6               | 90.8        | 72.8                | 75.9        | 74.1                  | 77.3        |
| AimCLR++ [6] (Journal PR 24)      | 80.9               | -           | 85.4               | -           | 70.1                | -           | 71.2                  | -           |
| Eq-Contrast [15] (IEEE TIP 24)    | 83.9               | -           | 90.3               | -           | 75.7                | -           | 77.2                  | -           |
| ViA [32] (IJCV 24)                | 78.1               | -           | 85.8               | -           | 69.2                | -           | 66.9                  | -           |
| SCD-Net [28] (AAAI 24)            | 86.6               | 87.3        | 91.7               | -           | 76.9                | -           | 80.1                  | -           |
| USDRL [27] (AAAI 25)              | 84.2               | 87.1        | 90.8               | 93.2        | 76.0                | 79.3        | 76.9                  | 80.6        |
| Heterogeneous [24] (CVPR 25)      | 80.2               | -           | 88.0               | -           | 70.7                | -           | 73.5                  | -           |
| <b>TambI (Our)</b>                | <b>87.1</b>        | <b>88.0</b> | <b>92.4</b>        | <b>93.3</b> | <b>79.0</b>         | <b>80.3</b> | <b>80.6</b>           | <b>81.5</b> |



# Result [Action recognition]: 2D Real-World datasets

| Methods                          | Linear Evaluation      |                        | Transfer Learning Evaluation<br>(After Pre-trained on Posetics) |                        |                        | Semi-supervised Evaluation |                        |                        |                        |
|----------------------------------|------------------------|------------------------|-----------------------------------------------------------------|------------------------|------------------------|----------------------------|------------------------|------------------------|------------------------|
|                                  | Posetics               |                        | SmartHome                                                       |                        | PennAction             | SmartHome CS (%)           |                        | PennAction             |                        |
|                                  | Top-1 (%)              | Top-5 (%)              | CS (%)                                                          | CV2 (%)                | Top-1 (%)              | (5%)                       | (10%)                  | (5%)                   | (10%)                  |
| OR-VPE [39] (FG 21)              | 14.6                   | 31.2                   | 42.7                                                            | 32.4                   | 78.5                   | -                          | -                      | -                      | -                      |
| 3s-CrosSCLR [16] (CVPR 21)       | 18.8                   | 38.1                   | -                                                               | -                      | -                      | -                          | -                      | -                      | -                      |
| AimCLR [30] (AAAI 22)            | 19.2                   | 39.3                   | 46.6                                                            | 48.3                   | -                      | -                          | -                      | -                      | -                      |
| CMD [22] (ECCV 22)               | 20.4                   | 40.5                   | 49.0                                                            | 52.5                   | 89.4                   | -                          | -                      | -                      | -                      |
| HiCLR [2] (AAAI 23)              | 20.1                   | 39.9                   | 49.1                                                            | 52.3                   | 88.7                   | -                          | -                      | -                      | -                      |
| HiCo [9] (AAAI 23)               | 21.3                   | 42.1                   | <u>54.3</u>                                                     | <u>54.8</u>            | 87.6                   | 34.5                       | <u>46.0</u>            | 57.2                   | 74.5                   |
| PCM <sup>3</sup> [42] (ACMMM 23) | 20.0                   | 40.3                   | 45.3                                                            | 46.8                   | 85.6                   | 23.1                       | 30.1                   | 46.3                   | 60.9                   |
| ViA [38] (IJCV 24)               | 20.7                   | 40.1                   | 49.5                                                            | 52.6                   | <u>90.2</u>            | <u>38.6</u>                | 45.3                   | <u>65.8</u>            | <u>85.2</u>            |
| USDRL [34] (AAAI 25)             | <u>25.9</u>            | <u>48.6</u>            | <u>54.3</u>                                                     | 53.9                   | 89.6                   | 37.1                       | 43.6                   | 58.1                   | 70.3                   |
| <b>Tambl (Ours)</b>              | <b>28.0</b><br>(+2.1%) | <b>51.0</b><br>(+2.4%) | <b>54.8</b><br>(+0.5%)                                          | <b>57.3</b><br>(+2.5%) | <b>91.0</b><br>(+0.8%) | <b>39.6</b><br>(+1.0%)     | <b>48.5</b><br>(+2.5%) | <b>69.2</b><br>(+3.4%) | <b>87.4</b><br>(+2.2%) |





# Result [Transfer Learning (left) and Semi-Supervised (right)]

| Method               | Transfer to PKU-MMD II |       |
|----------------------|------------------------|-------|
|                      | NTU-RGB+D 120          | CS(%) |
| HiCo-Transformer [9] | 55.4                   |       |
| CrosSCLR-B [16]      | 52.8                   |       |
| CMD [22]             | 57.0                   |       |
| UmURL-3 [29]         | 58.5                   |       |
| SCD-Net [35]         | 64.0*                  |       |
| Heterogeneous [31]   | 63.1                   |       |
| <b>TambI (Our)</b>   | <b>65.1</b>            |       |

| Method                | CS(%)       |             |             | CV(%)       |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                       | 1%          | 5%          | 10%         | 1%          | 5%          | 10%         |
| HiCLR [2]             | 39.1        | 63.3        | 70.7        | 42.9        | 68.3        | 74.8        |
| HiCo [9]              | 54.4        | 54.4        | 73.0        | 54.8        | 54.8        | 78.3        |
| PCM <sup>3</sup> [42] | 53.8        | -           | 77.1        | 53.1        | -           | 82.8        |
| UmURL [29]            | 58.1        | 72.5        | -           | 58.3        | 76.8        | -           |
| SCD-Net [35]          | 69.1        | -           | 82.2        | 66.8        | -           | 85.8        |
| USDRL (STTR) [34]     | 55.0        | -           | 76.1        | 59.1        | -           | 82.0        |
| USDRL (DSTE) [34]     | 57.3        | -           | 80.2        | 60.7        | -           | 84.0        |
| Heterogeneous [31]    | 55.0        | 76.3        | -           | 55.0        | 79.1        | -           |
| <b>TambI (Ours)</b>   | <b>69.2</b> | <b>79.9</b> | <b>83.0</b> | <b>68.7</b> | <b>83.7</b> | <b>87.3</b> |

NTU-RGB+D 60 dataset



# Result [Action Detection (left) and Action Prediction (right)]

| Method              | mAP@tIoU (%) |             |             |
|---------------------|--------------|-------------|-------------|
|                     | 0.1          | 0.3         | 0.5         |
| HiCo-Transf. [9]    | 32.5         | 31.8        | 28.6        |
| HiCo-LSTM [9]       | 39.2         | 37.2        | 32.0        |
| HiCo-GRU [9]        | 50.1         | 48.6        | 44.3        |
| LAC [40]            | 55.2         | -           | -           |
| SCDNet [35]         | 65.6         | 64.7        | 58.5        |
| <b>TambI (Ours)</b> | <b>72.6</b>  | <b>71.7</b> | <b>64.6</b> |

PKU-I dataset,  
Pretrained on NTU-RGB+D 60 CS

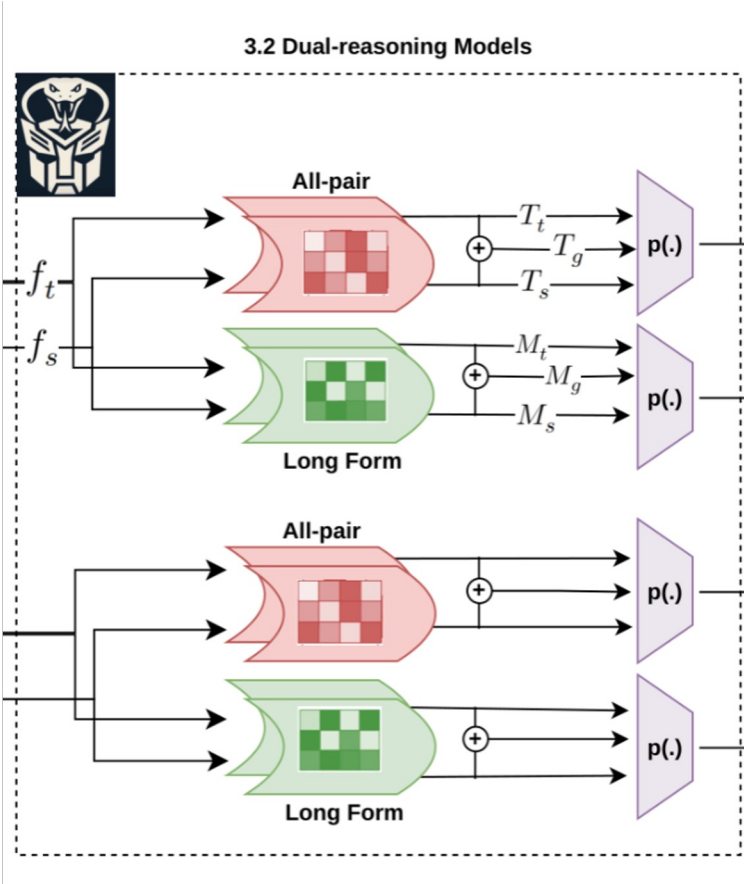
| Method                | 70%         | 80%         | 90%         | 100%        |
|-----------------------|-------------|-------------|-------------|-------------|
| <i>SmartHome-CV2</i>  |             |             |             |             |
| HiCo [5]              | 38.2        | 42.9        | 45.3        | 54.8        |
| PCM <sup>3</sup> [35] | 21.2        | 24.4        | 29.3        | 46.8        |
| USDRL (DSTE) [27]     | 48.2        | 51.2        | 52.5        | 53.9        |
| <b>TambI (Ours)</b>   | <b>49.5</b> | <b>53.3</b> | <b>55.4</b> | <b>57.3</b> |
| <i>Penn Action</i>    |             |             |             |             |
| HiCo [5]              | 77.8        | 80.6        | 83.4        | 87.6        |
| PCM <sup>3</sup> [35] | 53.4        | 56.7        | 62.4        | 85.7        |
| USDRL (DSTE) [27]     | 85.7        | 86.9        | 87.9        | 89.6        |
| <b>TambI (Ours)</b>   | <b>86.3</b> | <b>87.4</b> | <b>89.3</b> | <b>91.0</b> |
| <i>Posetics</i>       |             |             |             |             |
| HiCo [5]              | 18.5        | 19.9        | 20.5        | 21.3        |
| PCM <sup>3</sup> [35] | 17.1        | 18.1        | 18.7        | 20.0        |
| USDRL (DSTE) [27]     | 24.2        | 24.8        | 25.4        | 25.9        |
| <b>TambI (Ours)</b>   | <b>25.6</b> | <b>26.9</b> | <b>27.5</b> | <b>28.0</b> |



# Ablation Study-1

| Temporal                         | Spatial     | ICL | MiCo | NTU60 CV    |
|----------------------------------|-------------|-----|------|-------------|
| Single Reasoning                 |             |     |      |             |
| Trans                            | Trans       | ×   | ✓    | 91.7        |
| Mamba                            | Mamba       | ×   | ✓    | 83.9        |
| Trans                            | Mamba       | ×   | ✓    | 70.1        |
| Mamba                            | Trans       | ×   | ✓    | 85.2        |
| Dual Reasoning (Branch-specific) |             |     |      |             |
| Trans-Trans                      | Mamba-Mamba | ✓   | ✓    | 72.4        |
| Mamba-Mamba                      | Trans-Trans | ✓   | ✓    | 86.9        |
| Dual Reasoning (Ours)            |             |     |      |             |
| Trans-Mamba                      | Trans-Mamba | ✓   | ✓    | <b>92.4</b> |

Table 4.Dual Reasoning Design

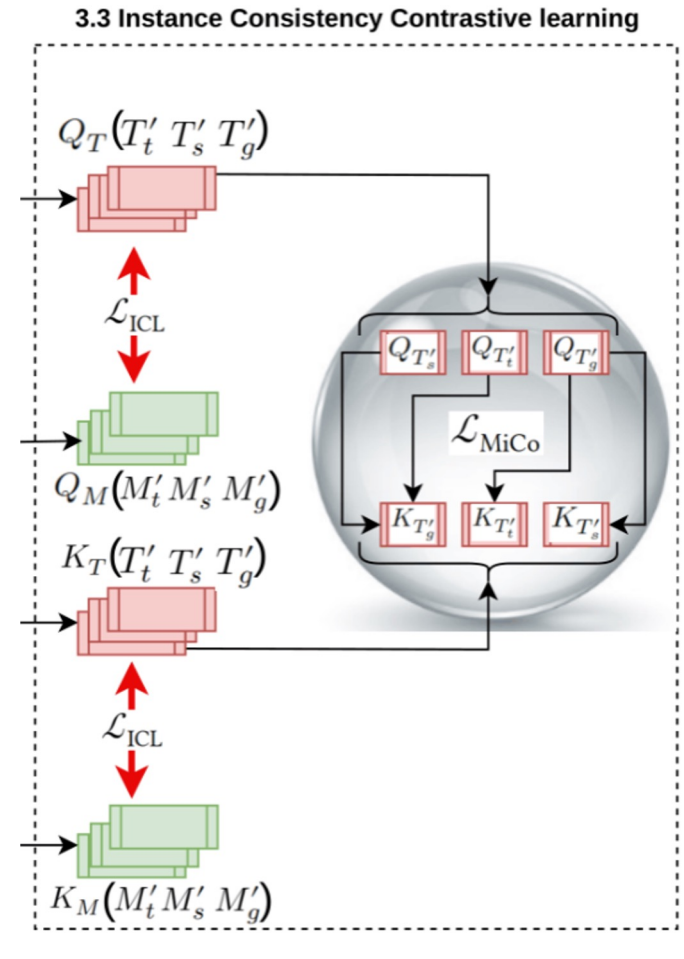




# Ablation Study-2

| Index | MiCo_loss<br>(Contrastive) | ICL<br>(Trans - Mamba) | NTU-RGB+D 60<br>CS(%) | CV(%)       |
|-------|----------------------------|------------------------|-----------------------|-------------|
| A0:   | Transformer                | -                      | 86.6                  | 91.7        |
| A1:   | Mamba                      | -                      | 83.0                  | 83.9        |
| A2:   | Both                       | -                      | 82.3                  | 81.0        |
| A3:   | Mamba                      | KL                     | 85.0                  | 88.9        |
| A4:   | Mamba                      | MSE                    | 84.8                  | 90.1        |
| A5:   | Transformer                | KL                     | 86.3                  | 91.2        |
| A6:   | Transformer                | MSE                    | <b>87.1</b>           | <b>92.4</b> |

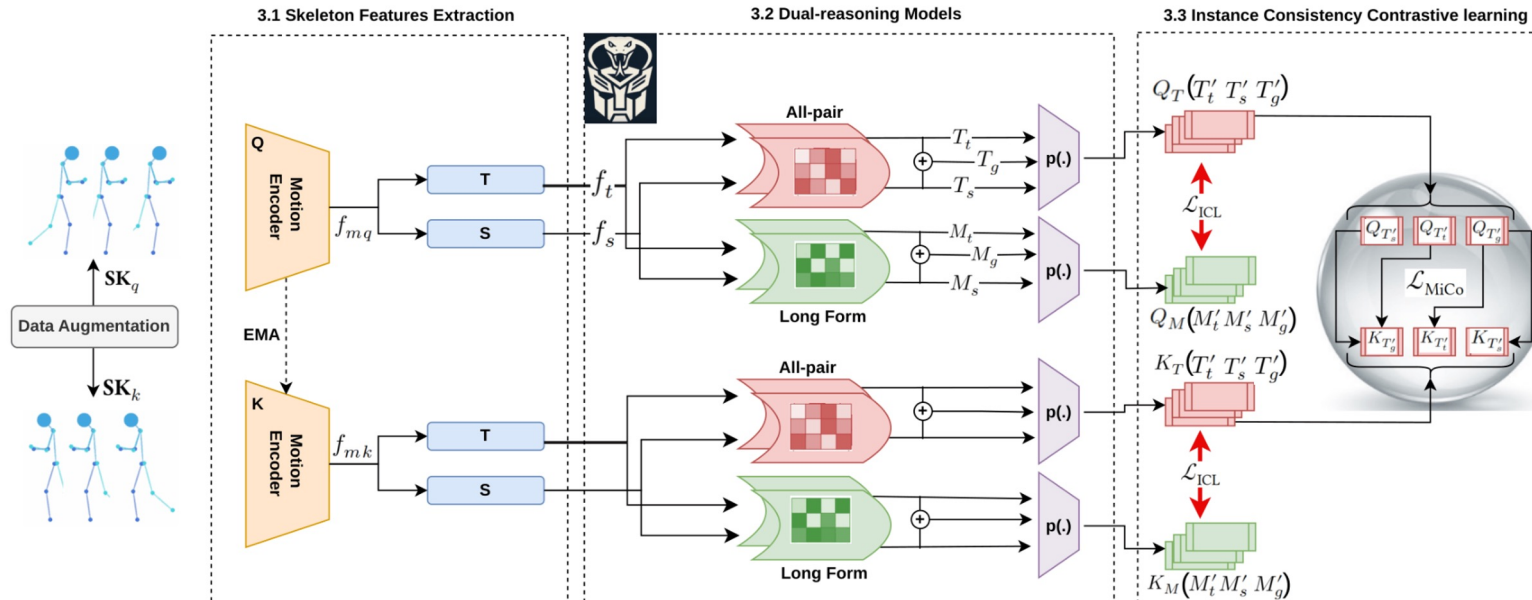
Table 6. Impact of ICL on MiCo.



# Ablation Study-3

| Method              | Decoupling | Reasoning  | Loss | NTU60 CV    | NTU120 CS   |
|---------------------|------------|------------|------|-------------|-------------|
| Baseline            | ×          | ×          | NCE  | 68.2        | 56.3        |
| HiCo [9]            | ✓          | ×          | NCE  | 88.6        | 72.8        |
| SCD-Net [35]        | ✓          | ✓ (Single) | NCE  | 91.7        | 76.9        |
| TambI (Ours)        | ✓          | ×          | NCE  | 88.9        | 76.2        |
| TambI (Ours)        | ×          | ✓ (Dual)   | ICCo | 86.3        | 73.8        |
| TambI (Ours)        | ✓          | ✓ (Dual)   | NCE  | 90.2        | 77.1        |
| <b>TambI (Ours)</b> | ✓          | ✓ (Dual)   | ICCo | <b>92.4</b> | <b>79.0</b> |

Table 5. Performance Comparison of Different Methods



# Deeper Insights and Analysis

| Activity                       | Gain from Tambl |
|--------------------------------|-----------------|
| <i>Long-Duration Activity</i>  |                 |
| Use tablet                     | +33.3           |
| Eat at table                   | +28.0           |
| Read book                      | +23.9           |
| <i>Short-Duration Activity</i> |                 |
| Sit down                       | +19.1           |
| Walk                           | +11.6           |

Table 6: Gain from Tambl (Transformer + Mamba) w.r.t. Transformer on SmartHome (Mean per Class).



Tambl

Vs.



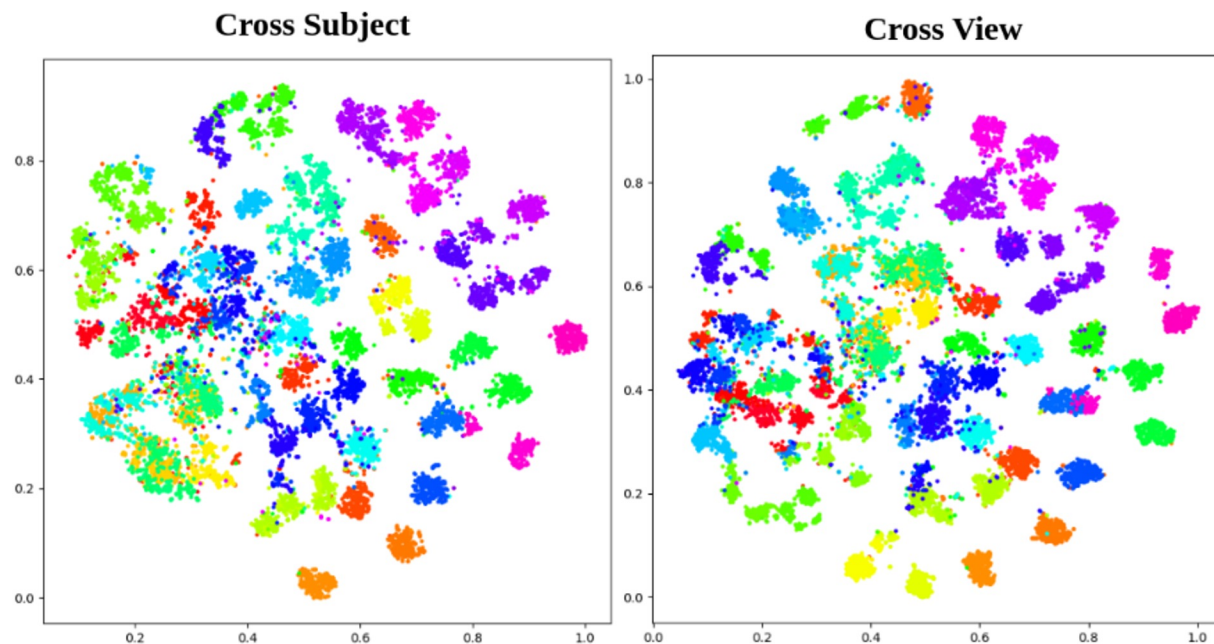
SCDNet



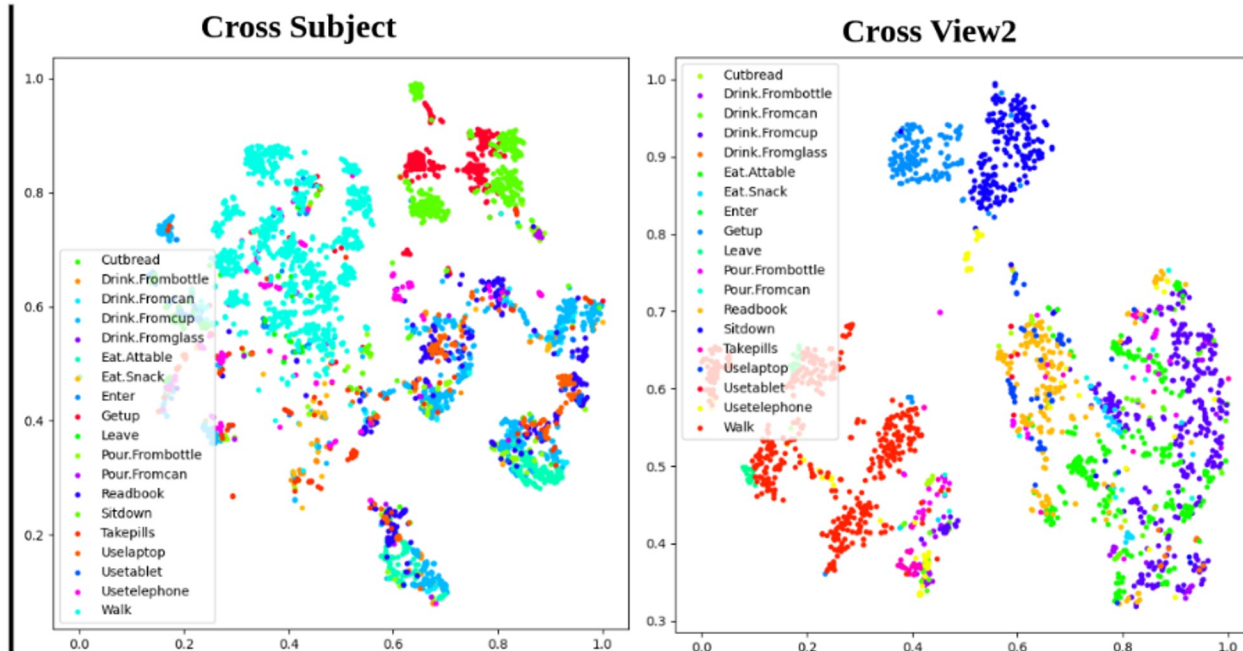


# Qualitative Results

a) Lab setting dataset [NTU RGB+D 60]



b) Real world dataset [SmartHome]



# Contributions and Conclusions

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- We Introduced a novel self-supervised methods (Tambl), that showcases superior generalizability and transferability to various scenarios of skeleton-based action understanding.
- Improve Skeleton representation via Dual Reasoning models [Transformer and Mamba].
- **Tambl:** A new indirect contrastive enhancement using dual objective learning that first aligns the cross-reasoning cues to enrich positive samples and then reinforces discrimination in the contrastive space.
- **SOTA Efficiency:** Outperforms all baselines on 6 benchmarks and **generalize across diverse scenarios**, including general activities, daily living tasks, and sports.



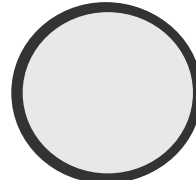
## MoVie: Broaden Your Views for Human Action Detection



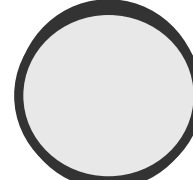
Di Yang



Mahmoud ALI



Xuanlong Yu



Xi SHEN



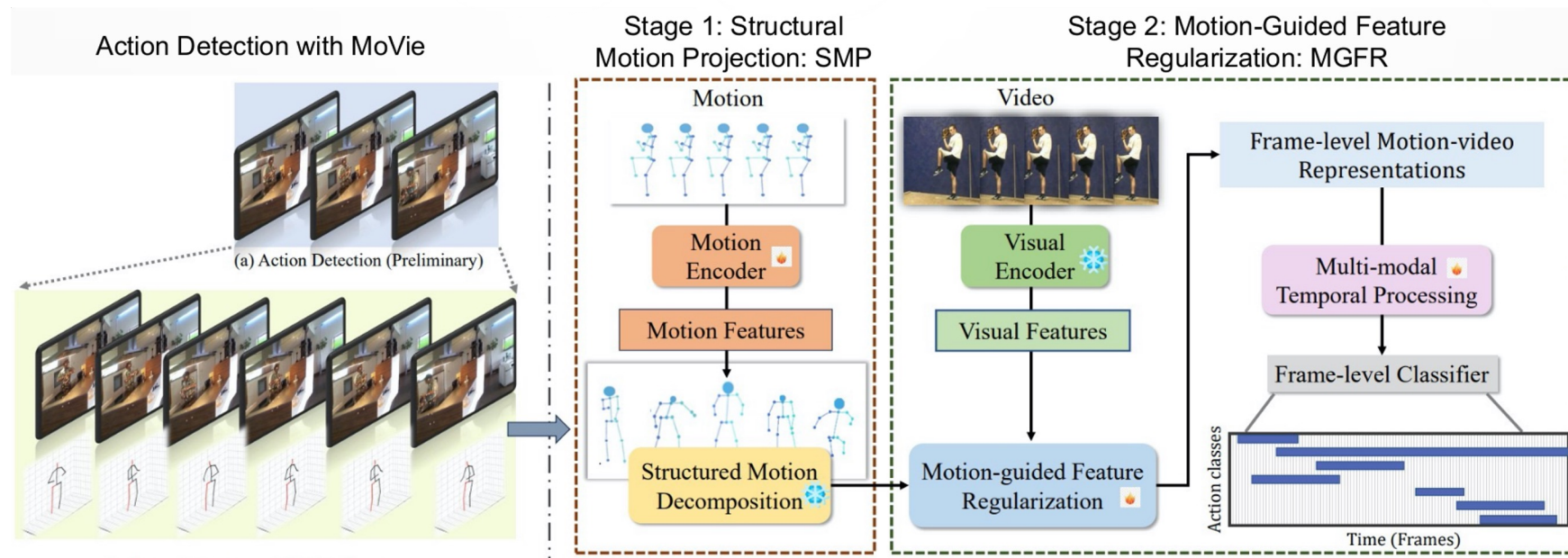
Quan Kong



Gianpiero Francesca

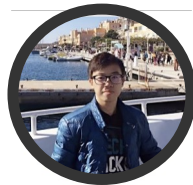


Francois Bremond





## T-MOR: Learning Motion-Aware Skeleton Representations for Human Action Recognition



Di Yang



Mahmoud Ali



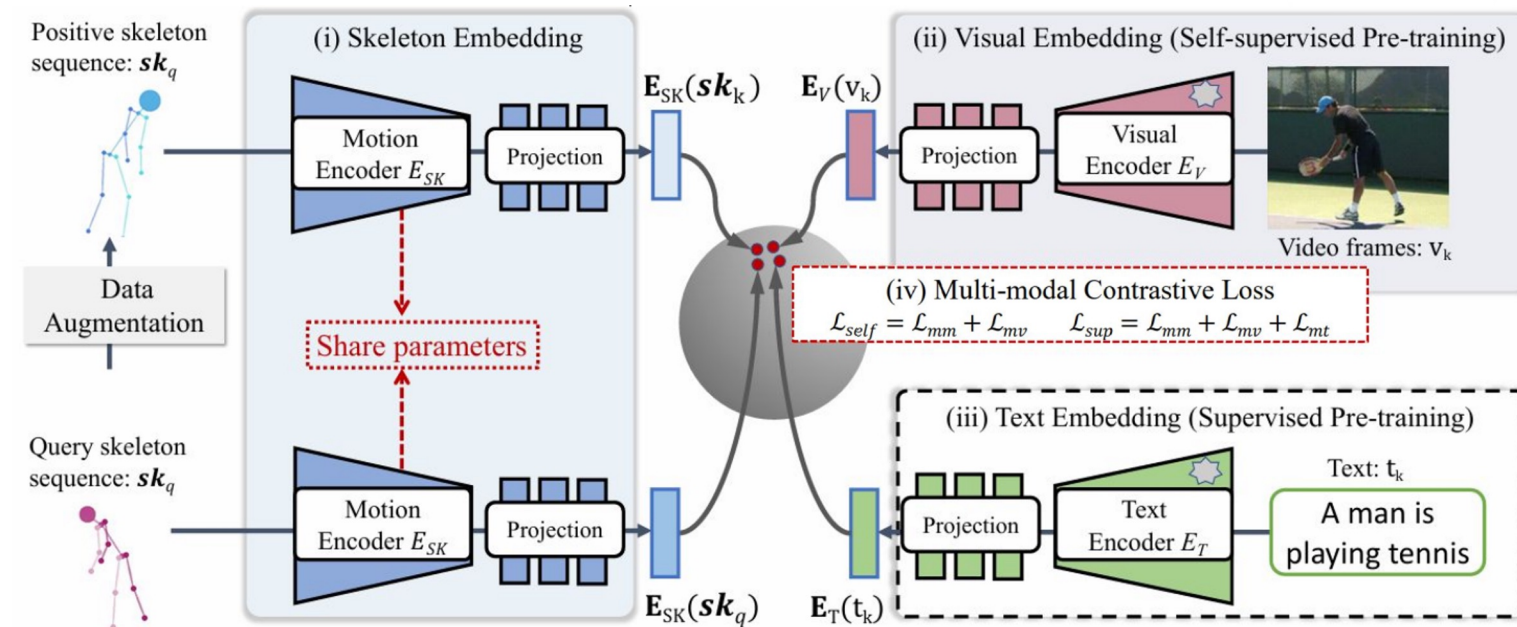
Quan Kong



Gianpiero Francesca



Francois Bremond



# Thanks



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**Di Yang\***



**Quan Kong**



**Gianpiero Francesca**



**François Brémont**



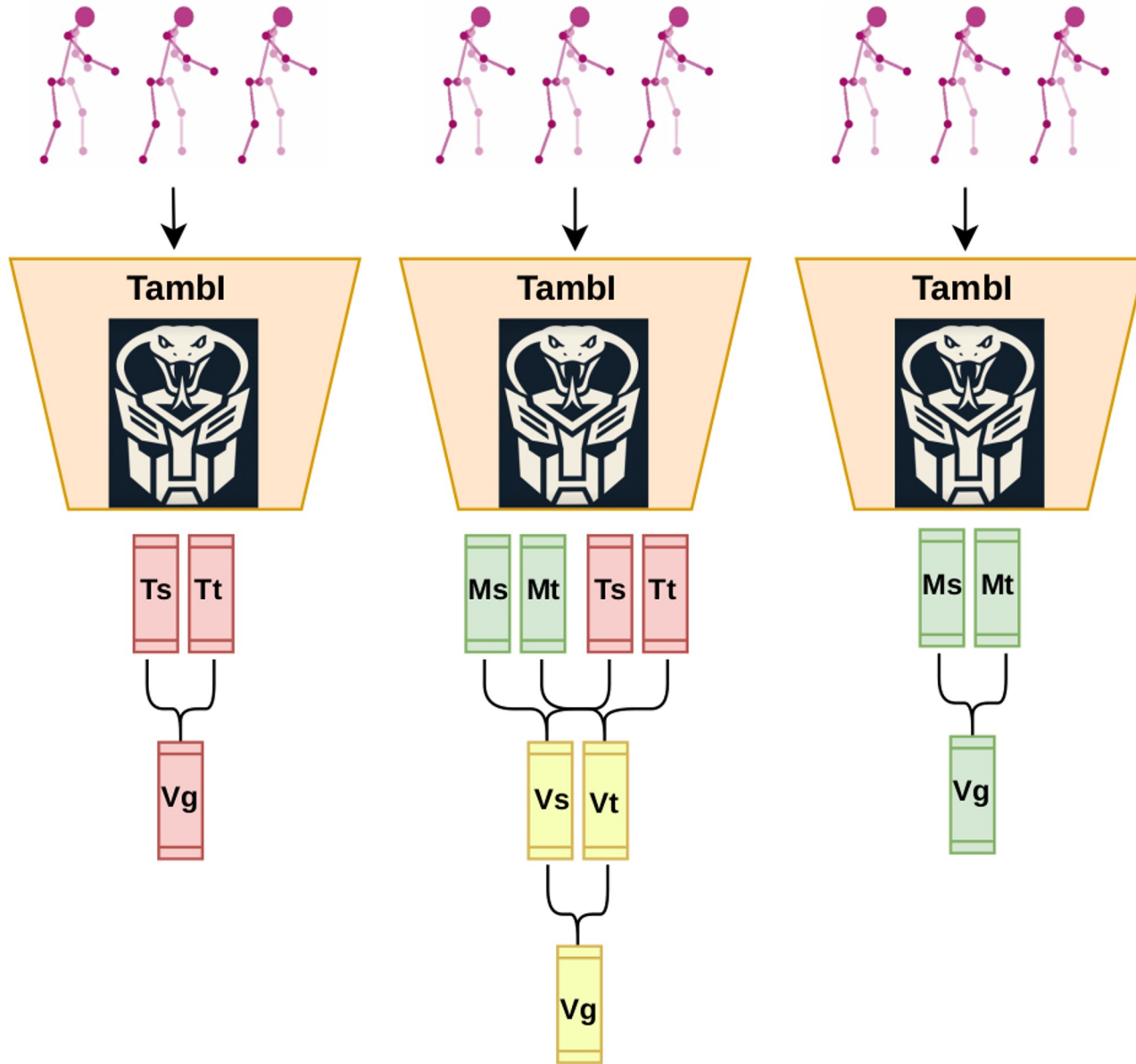
| Exp      | Mamba Config |            |        | NTU-RGB+D 60 |             |
|----------|--------------|------------|--------|--------------|-------------|
|          | $d_{state}$  | $d_{conv}$ | expand | CS(%)        | CV(%)       |
| Baseline | -            | -          | -      | <b>86.57</b> | <b>91.7</b> |
| Our      | 16           | 4          | 2      | 86.5         | 91.8        |
| Our      | 32           | 4          | 2      | 86.2         | 85.9        |
| Our      | 64           | 4          | 2      | <b>87.1</b>  | <b>92.3</b> |
| Our      | 128          | 4          | 2      | 85.3         | 90.0        |



| GCN Layer | Trans Layer | Trans Head | Top-1 (%)   |
|-----------|-------------|------------|-------------|
| 4         | 1           | 8          | 86.6        |
| 3         | 2           | 8          | 86.4        |
| 3         | 1           | 16         | 85.7        |
| 3         | 1           | 8          | <b>87.1</b> |







(a) Transformer

(b) Both

(c) Mamba

| Inference   | NTU-RGB+D 60 |             | NTU-RGB+D 120 |             |
|-------------|--------------|-------------|---------------|-------------|
|             | CS(%)        | CV(%)       | CS(%)         | Cset(%)     |
| Both        | <b>87.2</b>  | 92.1        | 78.9          | <b>80.7</b> |
| Mamba       | 82.8         | 85.7        | 75.4          | 77.0        |
| Transformer | 87.1         | <b>92.4</b> | <b>79.0</b>   | 80.6        |

